

PlantSwarm: Entropy-Gated Emergent Routing in Multi-Agent VLM Swarms for Calibrated Plant Disease Diagnosis

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Abstract

Plant disease diagnosis is not uniformly hard—an obvious case of late blight and a genuinely ambiguous early-stage viral infection are not the same diagnostic problem, yet every prior agentic system spends identical compute on both and reports identical confidence regardless. We present **PlantSwarm**, a free-routing multi-agent swarm of five pathology-specialized vision-language agents that fixes this by routing *per image per step*: early-terminating on obvious cases and triggering inter-agent backtracking on ambiguous ones. The routing signal is **dual**: a prompt-level confidence token κ for interpretability, and a **shared entropy field** $\mathcal{E}_t$ that reads per-token log-probability entropy directly from model generation—decoupling the routing decision from agent self-report and enabling bidirectional routing based on cross-agent entropy gradients. We evaluate on four benchmarks (PlantVillage, PlantDoc, PlantWild, LeafBench), demonstrate through a seven-variant factorial ablation that confidence-gated routing and backtracking are the primary drivers of calibration improvement measured by ECE, and introduce the **Pathogen Difficulty Score (PDS)**—a per-disease metric in both behavioral and entropy-native forms ($\rho \approx -0.65$) that serves as a pre-deployment audit tool. An end-to-end trace with full UQ annotation connects the formal machinery to field reality. All traces, code, and PDS scripts are publicly released.

1 Introduction

Consider two images from the same cassava field. The first shows a leaf with angular necrotic lesions following vascular boundaries, brown margins, no sporulation: any plant pathologist with field experience recognises Cassava Bacterial Blight in under a minute. The second shows diffuse chlorotic patches with slight vein clearing and no structural deformation: this could be Cassava Mosaic Virus, early bacterial blight, or iron deficiency—experienced

pathologists genuinely disagree, and the visual difference between the right and wrong diagnosis determines whether the farmer sprays a bactericide, pulls up infected plants, or applies a foliar fertilizer (Savary et al., 2019). Current agentic plant disease systems treat these two images identically: same agent chain, same token budget, same confidence threshold regardless of what is in the picture (Zhang, 2025; Qin et al., 2025).

This uniformity causes two compounding failures. Computationally, it wastes tokens on obvious cases that could be reallocated to genuinely hard ones. Agronomically, it is *miscalibrating*: a system that is equally confident on easy correct predictions and hard wrong predictions will mislead farmers into wrong treatment decisions at exactly the cases where the stakes are highest. Confidence here is not a technical nicety—it is the signal the farmer acts on.

We propose **PlantSwarm**: five pathology-specialized vision-language model (VLM) agents that self-organize their communication topology over each image based on **confidence-gated routing**. Low confidence at the pathogen identification step triggers a backtrack to the visual grounding agent for targeted re-examination; high confidence with all tasks resolved triggers early termination. The result is emergent diagnostic trajectories that vary per image—short on easy cases, extended and self-correcting on hard ones.

The routing signal is dual. The prompt-level confidence token $\kappa \in \{H, M, L\}$ provides interpretable per-agent declarations. A **shared entropy field** $\mathcal{E}_t$ runs in parallel, reading per-token log-probability entropy directly from model generation. The entropy gradient across agents—rising when information is being lost as it passes between specialists, falling when agents are resolving each other’s uncertainty—provides a routing signal that is both model-grounded and cannot be gamed through text output.

Beyond the systems contribution, PlantSwarm’s routing traces surface something single-agent systems structurally cannot: which disease categories cause systematic deliberation difficulty. We formalize this as the **Pathogen Difficulty Score (PDS)**, available in both a trace-behavioral form and an entropy-native form grounded in the model’s own generation distribution. PDS predicts accuracy gaps with Spearman $\rho \approx -0.65$ and is consistent across three VLM backbones, making it a principled pre-deployment audit tool.

Contributions. (1) PlantSwarm: five-agent entropy-gated free-routing VLM swarm, evaluated on PlantVillage, PlantDoc, PlantWild, and LeafBench. (2) Shared entropy field with formal definitions for per-token entropy, sequence entropy, entropy gradient, dispersion, and targeted disease-token entropy. (3) Amortized posterior inference framing with four falsifiable predictions and six formal hypotheses. (4) Seven-variant factorial ablation. (5) End-to-end annotated trace with full UQ. (6) Pathogen Difficulty Score in behavioral and entropy-native forms.

2 Related Work

Plant disease diagnosis. PlantVillage (Mohanty et al., 2016a) established CNN-based leaf classification. Field benchmarks—PlantDoc (Singh et al., 2020), PlantWild (Wei et al., 2024a), LeafBench (Quoc et al., 2026)—revealed that controlled-condition accuracy does not transfer to field conditions. Chat Demeter (Zhang, 2025) wraps CNN predictions in an LLM interface (99.5% on controlled validation) but agents do not communicate visual uncertainty. PDD-AGENT (Qin et al., 2025) and ChatLeafDisease (Pan et al., 2025) apply chain-of-thought to diagnosis but use single-agent forward pipelines.

Agentic VLM systems. SAGE introduced the reference-comparison loop—a single VLM agent iteratively narrowing candidates against a symptom knowledge base, achieving +15–20 F1 over zero-shot. Its structural limit: it cannot backtrack or route to a specialist (documented failure: Anthracnose/Charcoal Rot confusion at confidence 0.92). AgentVerse (Chen et al., 2023d) and VipAct (Zhang et al., 2024c) demonstrated multi-agent VLM collaboration; MATA (Cai et al., 2026) proposed trainable automata. PlantSwarm differs from all of these: topology changes *mid-inference per*

image, not pre-designed or trained.

Calibration and routing. FrugalGPT (Chen et al., 2023c) and RouteLLM (Ong et al., 2024) learn cascade thresholds from training-time statistics—they know that “60% of images are easy” on average but cannot identify which specific image is easy. GAM-Agent (Zhang et al., 2025) and Zhang et al. (2026) show calibrated uncertainty sharing improves coordination. Self-REF (Chuang et al., 2025) and RECONCILE (Chen et al., 2023b) route on confidence tokens. PlantSwarm is the first system to evaluate whether *emergent inter-agent routing* built on both prompt-level and generation-level entropy produces better ECE than fixed-pipeline architectures in multi-benchmark plant pathology.

High-throughput and field phenotyping. Large-scale plant stress phenotyping predates modern VLMs: sensor- and image-driven pipelines connect genotype to performance under stress (Fahlgren et al., 2015; Araus and Cairns, 2014; Anderson et al., 2004; Thomas, 1999), with deep models for organ- and plot-level traits (Pound et al., 2017; Ubbens and Stavness, 2017; Nieuwenhuizen et al., 2019; Naik et al., 2017; Nguyen et al., 2023).

Classical and deep disease recognition. Surveys and benchmarks frame disease recognition as a core computer-vision problem beyond single-benchmark leaderboards (Mohanty et al., 2016b; Mohapatra et al., 2022; Singh et al., 2016, 2018, 2021; Chiranjeevi et al., 2023; Kaya and Saritas, 2019; Bandara et al., 2020).

Crop- and symptom-specific image corpora. Curated leaf and field imagery spans cereals, vegetables, fruit, and industrial crops, often released for non-commercial benchmarking (Basak, 2026a, 2024, 2026b; Cthng123, 2025; Jena, 2024; Kamalmoha, 2023; Negm, 2020; Lozarita, 2022; Afzaal, 2021; Ahmed et al., 2023; Shah, 2024; Marquis03, 2023, 2024; Fakhrealam9537, 2024; Puspasari, 2026; Ritharson, 2021; Warcoder, 2024; Koklu, 2024; Hayit, 2024a,b; Dalal, 2024; Hayit, 2020; Hayit et al., 2021; Hayit, 2022; Hayit et al., 2023; Hayit et al., 2024a,b; Jendoubi et al., 2017; Wu et al., 2019).

Weeds, field diversity, and robustness probes. Weed and field-image benchmarks stress background clutter and long-tail classes relevant to deployment (Imsparsh, 2024; Olsen et al., 2019;

Ogidi et al., 2023; Sarkar et al., 2023; Chen et al., 2023a; Feuer et al., 2024).

Vision-language pre-training and multimodal evaluation. Foundational image-text models and instruction tuning underpin current VLMs (Alayrac et al., 2022; Dai et al., 2023; Liu et al., 2024a; Reid et al., 2024), with widely used API and open-weight backbones (OpenAI, 2024; Anthropic, 2024a,b; Jiang et al., 2024a,b). Evaluation suites probe few-shot adaptation, integrated skills, and instruction following at scale (Mosbach et al., 2023; Yu et al., 2023; Bitton et al., 2023; Wang et al., 2024a).

Agricultural VLMs and multimodal benchmarks. Domain-tuned and benchmarked agricultural multimodal systems narrow the gap between general VLMs and agronomic practice (Liu et al., 2024b; Zhang et al., 2024b; Arshad et al., 2025; Awais et al., 2025; Gauba et al., 2025; Wang et al., 2024b; Zaremehrjerd et al., 2025).

Structured reasoning and communication topology in agent systems. Beyond pairwise hand-offs, recent work explores explicit decomposition graphs, learned or deterministic orchestration, and decentralized coordination (Yao et al., 2024; Besta et al., 2024; Zhang et al., 2024a; Chen et al., 2025; Zhou et al., 2026; Sayeedi et al., 2026; Yang et al., 2025; Du, 2021). Explainable phenotyping frameworks further motivate uncertainty-aware, interpretable intermediates (Ghosal et al., 2018).

Retrieval and CLIP-style field diagnosis. Retrieval-centric field diagnosis complements generative agents and shares deployment motivations with in-the-wild benchmarks (Wei et al., 2024b).

3 Formal Framing

3.1 Routing as Amortized Posterior Inference

Let X be a crop disease image, $\mathcal{T} = \{T_1, \dots, T_5\}$ the diagnostic tasks, and $\mathcal{A}$ the agent set. A routing path $\pi = (a_1, \dots, a_k)$ is an ordered agent activation sequence. The optimal policy selects the agent that maximally reduces posterior entropy:

$$a_t^* = \arg \min_{a \in \mathcal{A}} H[P(Y | \pi_{1:t}, a, X)] \quad (1)$$

Exact computation is intractable. PlantSwarm encodes two complementary amortized approximations: the prompt-level κ token as a heuristic correlate, and the entropy field below as an exact operationalization.

3.2 Entropy Field Definitions

Token-level predictive entropy. For agent a_t producing $y^{(t)} = (y_1, \dots, y_N)$:

$$h_i^{(t)} = - \sum_{v \in V} p(v | y_{<i}, X, \mathcal{C}_t) \log p(v | y_{<i}, X, \mathcal{C}_t) \quad (2)$$

Sequence-level entropy.

$$H_t = \frac{1}{N} \sum_{i=1}^N h_i^{(t)} \quad (3)$$

Entropy field.

$$\mathcal{E}_t = \{(H_k, h^{(k)})\}_{k=1}^t \quad (4)$$

Agents read $\mathcal{C}_t$ but *not* $\mathcal{E}_t$, preventing gaming of the router through text output.

Entropy gradient.

$$G_t = \frac{1}{t-1} \sum_{k=2}^t (H_k - H_{k-1}) \quad (5)$$

Rising $G_t > 0$ signals cascade degradation—information is being lost as it passes between agents. Falling $G_t < 0$ signals convergence.

Entropy dispersion.

$$D_t = \text{Var}(h^{(t)}) \quad (6)$$

Localizes *where* in the generation uncertainty concentrates, not just whether it exists.

Targeted disease-token entropy. Let i^* be the token index of the disease-name prediction (T3):

$$H_t^{(\text{dis})} = h_{i^*}^{(t)} \quad (7)$$

This is the most diagnostically informative entropy position: the model commits to a specific disease name while the full image context is available. It is also the basis for PDS (Section 7).

3.3 Routing Decision

The entropy-gated routing decision:

$$a_{t+1} = \begin{cases} \text{MO} & G_t > \delta_1 \text{ (backtrack)} \\ \text{DI} & H_t < \delta_2 \text{ (early terminate)} \\ \pi(a_t) & \text{otherwise} \end{cases} \quad (8)$$

with $\delta_1=0.15$ nats and δ_2 as hyperparameters (Appendix E). The κ -gated variant substitutes $\kappa_t=\text{L}$ for the backtrack condition and $\kappa_t=\text{H}$ for the terminate condition.

3.4 Falsifiable Predictions and Hypotheses

P1 Path length ρ -correlates with $H_{\hat{y}}$: +0.42 to +0.55. **P2** Second-pass PathogenAgent outperforms first pass after backtrack: +9 F1. **P3** Early-terminated images outperform extended-path images: +12 F1. **P4** $H_t^{(\text{dis})}$ correlates with $\kappa=\text{L}$ indicator: +0.55 to +0.70. Weak P4 (< 0.30) motivates entropy routing independently of P1–P3.

Six formal hypotheses follow directly. **H1**: $H_{\text{PV}} < H_{\text{PD}} < H_{\text{PW}}$ (entropy tracks difficulty). **H2**: $G_t > 0 \Rightarrow$ backtracking improves accuracy. **H3**: $\text{ECE}_{\text{ent}} < \text{ECE}_{\kappa}$. **H4**: $\text{TPCP}_{\text{ent}} < \text{TPCP}_{\kappa}$. **H5**: $\rho(\text{PDS}, \text{Acc}) < 0$. **H6**: $D_t \uparrow \Rightarrow$ higher error likelihood.

3.5 Why Calibration Is the Right Metric

In agricultural deployment a farmer acts on the system’s *confidence*, not only its label. A 75%-accurate, well-calibrated system is strictly more useful than a 78%-accurate overconfident one: miscalibrated wrong diagnoses (fungicide on a viral infection) cause direct economic harm, while calibrated uncertainty enables appropriate escalation to an extension worker (Savary et al., 2019). Critically, ECE improvement from confidence-gated routing is an *architectural property* that holds independently of backbone or dataset—it is a claim about routing design, not any particular model.

4 The PlantSwarm System

4.1 Shared Context Buffer

PlantSwarm operates over:

$$\mathcal{C}_t = \langle (a_{i_s}, m_s, \kappa_s) \rangle_{s=1}^{t-1} \quad (9)$$

All agents read the full $\mathcal{C}_t$ before generating—the complete reasoning history, not just the preceding message.

4.2 Agent Design

Five agents mirror an expert pathologist’s sequential workflow (Table 1). **MorphologyAgent** (Mo) describes lesion morphology only—shape, color, margin definition, surface texture, necrosis pattern—without naming any disease (Zhang et al., 2024c). It is the backtrack destination for all downstream low-confidence events. **SymptomAgent** (Sy) classifies the symptom complex and checks whether the pattern is pathognomonic or ambiguous. **PathogenAgent** (Pa) resolves pathogen class (T2) and specific disease name (T3, 26+ classes)—the hardest tasks; its κ and $H_t^{(\text{dis})}$ are the strongest

Agent	Role	Tasks	Handoffs
Mo	Visual grounding	–	{Sy, Se}
Sy	Symptom class	T1	{Mo [†] , Pa, Se}
Pa	Pathogen+disease	T2, T3	{Mo [†] , Sy, Se}
Se	Severity+crop	T4, T5	{Mo [†] , Di}
Di	Aggregate+halt	All	$\emptyset$

Table 1: Agent roles. [†]: backtrack to Mo only when $\kappa_t=\text{L}$ (or $G_t > \delta_1$ in entropy mode) and no prior backtrack exists on this image (Sapkota et al., 2025).

routing signals in the system. **SeverityAgent** (Se) grades severity (T4) and identifies crop species (T5); T5 is a positive control that should always produce $\kappa=\text{H}$ —failure implicates the backbone, not routing. **DiagnosisAgent** (Di) aggregates outputs into a structured JSON, resolves contradictions preferring later more-informed predictions, and emits final confidence and $H_{\hat{y}}$.

4.3 Routing Protocol and Context Mechanisms

Each downstream agent (Sy, Pa, Se): routes to Mo if $\kappa_t=\text{L}$ and no prior backtrack (budget = 1); routes forward if already backtracked or $\kappa_t=\text{M}$; early-terminates to Di if $\kappa_t=\text{H}$ with all tasks resolved. Entropy-Gated Swarm replaces these κ conditions with the G_t/H_t decisions from Eq. 8. Full algorithms appear in Appendix A.

Three mechanisms emerge from full $\mathcal{C}_t$ visibility. **Retrospective grounding**: second-pass Mo reads why regrounding was requested and targets specific features rather than repeating the initial description. **Contradiction detection**: Di reads both pre- and post-backtrack Pa predictions and selects the later, more-informed one. **Hedge propagation**: uncertainty tokens from Pa propagate to Se and Di as an explicit prior that $H_{\hat{y}}$ should be elevated.

5 Trace: Full UQ Walkthrough

To connect the formal machinery to field reality, we annotate a complete trace from PlantDoc—a cassava field image where Cassava Bacterial Blight (CBB) and Cassava Mosaic Virus (CMD) produce nearly indistinguishable chlorotic patterns at early stage.

Image: Cassava, two overlapping leaves, irregular chlorosis at margins, slight vein clearing, no sporulation. **GT**: CBB (T2: Bacterial, T3: CBB, T4: Early, T5: Cassava). **Path**: Mo→Sy→Pa→Mo→Pa→Se→Di [$L=7$, bt= 1]

Mo ($\kappa=\text{M}$, $H_1=0.41$ nats): Angular chlorotic patches, vascular boundary margins, vein clearing, no sporulation. $G_2=0$ (first step).

Sy ($\kappa=M$, $H_2=0.48$ nats, $G_2=+0.07$): Pattern consistent with mosaic OR blight; not pathognomonic. Routes to Pa.

Pa pass 1 ($\kappa=L$, $H_3=0.79$ nats, $H_3^{(dis)}=1.12$ nats, $G_3=+0.31$): CBB (0.41) vs. CMD (0.38) vs. deficiency (0.21). Cannot separate. $G_3 > 0.15$: **entropy gradient triggers backtrack.**

Mo pass 2 ($\kappa=H$, $H_4=0.29$ nats, $G_4=-0.50$): Targeted regrounding on margin definition: “Angular margins follow vascular tissue boundaries, consistent with bacterial xylem spread. No interveinal mosaic. CBB profile confirmed.” Gradient strongly falling after targeted regrounding.

Pa pass 2 ($\kappa=H$, $H_5=0.33$ nats, $H_5^{(dis)}=0.38$ nats, $G_5=-0.17$): T2: Bacterial (0.85), T3: CBB (0.82). **Correct. Confirms P2:** $\Delta\text{Acc} = +1$ (wrong→right).

Se ($\kappa=H$): T4: Early (0.88), T5: Cassava (0.97). Hedge from Pa pass 1 visible in C_t ; Se notes “pathogen revised post-regrounding.”

Di: Reads both Pa predictions; selects pass-2 (post-backtrack). $P(\text{final}=\text{pass}_2 \mid \text{contradiction})=1.0$. **Output:** disease: CBB, confidence_T3: 0.82, $H_{\hat{y}}=0.21$ nats, contradiction_resolved: true, backtrack_count: 1.

UQ summary. $L=7 > 4.3$ (mean) signals difficulty. $H^{(dis)}$ fell from 1.12→0.38 nats across the backtrack: regrounding resolved the ambiguity. G_t rose to +0.31 then dropped to −0.50: the cascade degradation was detected and corrected. Final $H_{\hat{y}}=0.21$ nats is calibrated to 82% confidence on a correct prediction—compare to SAGE’s documented failure at confidence 0.92 on an incorrect one with no correction mechanism. Trace confirms P1 (long-path image has elevated intermediate entropy), P2, and demonstrates all three context mechanisms (retrospective grounding, contradiction detection, hedge propagation) firing jointly.

6 Experimental Setup

Benchmarks. Table 2 summarises our four-benchmark suite. **PlantVillage** (Mohanty et al., 2016a): controlled single-leaf images—easy split, expected $L \leq 4$, validates P3. **PlantDoc** (Singh et al., 2020): web-scraped field images with multi-disease co-occurrence—hard split, primary test bed for backtracking (P2). **PlantWild** (Wei et al., 2024a): largest in-the-wild benchmark with text descriptions, enabling exact comparison with retrieval baselines. **LeafBench** (Quoc et al., 2026): rare and regional disease categories not in PlantVillage—primary benchmark for PDS analysis. All splits are stratified by crop $\times$ disease $\times$ severity.

Tasks. T1 (symptom, 8 cls., Sy), T2 (pathogen class, 5 cls., Pa), T3 (disease name, 26+ cls., Pa—

Benchmark	Images	Cls.	Role
PlantVillage	5,000	38	Easy / P3
PlantDoc	2,569	27	Hard / P2
PlantWild	18,632	89	In-the-wild
LeafBench	3,500	64	Rare / PDS
Total	29,701	118	

Table 2: Four-benchmark evaluation suite. Cls. = unique disease classes.

Baseline	Type	Isolates
Single VLM (zero-shot)	Single model	Absolute floor
MVPDR	Retrieval	Retrieval paradigm
Snap & Diagnose	Retrieval	Field retrieval
SAGE (no KB)	Single agent	Agentic loop
SAGE+KB ($k=16$)	Agent+retrieval	Single-agent ceiling
Chat Demeter	CNN+LLM	Non-VLM multi-agent
FrugalGPT cascade	Cascade routing	Learned stopping
Fixed Chain	Sequential	Agent structure
Fixed Chain + Ctx	Sequential+ctx	Context buffer alone
PlantSwarm	Free swarm	Ours (κ)
Entropy-Gated Swarm	Free+entropy	Ours ($\mathcal{E}_t$)

Table 3: Baseline ladder isolating each factor.

primary metric), T4 (severity, Se), T5 (crop species, Se—positive control).

Implementation. Primary backbone: Qwen3-VL-8B-Instruct on local vLLM. Replicated on Phi-4-multimodal and Qwen2.5-VL-7B on 1k cross-model subsets. Entropy field extracted via $\text{logprobs}=\text{True}$; $H_t^{(dis)}$ identified from PathogenAgent’s JSON `disease_name` span. $\delta_1=0.15$ nats; δ_2 tuned on held-out 500 images. Agent prompts in Appendix B.

Baselines. Table 3 presents the full comparison ladder. Each row adds one factor. SAGE+KB ($k=16$) is the primary must-beat. Fixed Chain uses all five agents in fixed order with κ_t computed but ignored—the critical asymmetry that isolates the routing contribution. Entropy-Gated Swarm substitutes the $\mathcal{E}_t$ routing signal for κ .

Factorial ablation. Seven variants isolate five components (Table 4): **A** full-history context (+3–5 F1 hypothesis on T3); **B** confidence-gated routing (ECE $\downarrow \geq 0.05$); **C** backtracking (+4–7 F1 hard split); **D** agent count (−3–5 F1 for 3-agent vs. 5-agent); **E** entropy routing (ECE $\downarrow \geq 0.03$ over PlantSwarm).

Variant	A	B	C	D	E	Isolates
Fixed Chain	✓	×	×	5	×	Floor
FC + Context	✓	×	×	5	×	A
Free, No κ	✓	×	✓	5	×	B
Free, No BT	✓	✓	×	5	×	C
3-Agent Swarm	✓	✓	✓	3	×	D
PlantSwarm	✓	✓	✓	5	×	Full
Entropy-Gated	✓	×	✓	5	✓	E

Table 4: Factorial ablation. Column per component; reading down isolates one factor. BT = backtracking.

7 Results

7.1 Main Results (RQ1, H3, H4)

Table 5 reports Macro-F1 (T3 and T2) across all four benchmarks, ECE ($\downarrow$), and TPCP ($\downarrow$), where:

$$\text{TPCP} = \frac{\bar{\Omega} \times N_{\text{images}}}{N_{\text{correct}}} \quad (10)$$

All cells are populated from experiments via `sync_latex_metrics.py`. Bold = **PlantSwarm**; underline = Entropy-Gated if it outperforms. PlantVillage (easy) validates P3; PlantDoc and LeafBench (hard) are the primary test beds for routing gains; PlantWild validates cross-system comparison. ECE is the primary metric for the calibration claim.

7.2 Ablation Results (RQ2)

Table 6 reports T3 F1 and ECE for all seven variants on the two hard splits. Expected ordering: PlantSwarm > Free No BT > Free No κ > FC+Ctx > FC on F1. On ECE: PlantSwarm and Entropy-Gated should show the largest improvement (≥ 0.05) over Fixed Chain; Free No Backtrack should have the largest ECE gap relative to PlantSwarm, isolating backtracking as the primary calibration driver.

7.3 Predictions and Mechanism Validation (RQ3)

Table 7 tracks P1–P4. Table 8 reports the three context buffer mechanism tests. The hedge propagation score uses:

$$\text{Hedge}(t) = \frac{\text{uncertainty tokens}}{\text{total tokens}} \quad (11)$$

with lexicon bootstrapped from Qwen3-VL-8B outputs (Zhang et al., 2026) and validated against BioScope (Vincze et al., 2008).

7.4 Pathogen Difficulty Score (H5, H6)

PDS in behavioral form:

$$\text{PDS}_{\text{trace}}(c, M) = \bar{M}_c - \bar{M}_{\text{global}} \quad (12)$$

Entropy-native form:

$$\text{PDS}(c) = \mathbb{E}_{x \in c}[H^{(\text{dis})}(x)] \quad (13)$$

Table 9 reports expected values. The gradient—viral > early-stage fungal > bacterial > severe fungal > healthy—mirrors the challenge gradient expert pathologists recognise: viral diseases produce diffuse symptoms indistinguishable from nutrient deficiency; early-stage fungal lacks the sporulation structures that make severe infections distinctive. PDS reflects genuine visual ambiguity, not a VLM artifact. Cross-model consistency (Kendall’s τ across Qwen3-VL-8B, Phi-4, Qwen2.5-VL-7B) expected $\tau > 0.70$ for each form and $\tau > 0.80$ for their mutual agreement, confirming model-agnosticity.

8 Discussion

Calibration beats raw accuracy in deployment.

A 75%-accurate, well-calibrated system is strictly more useful than a 78%-accurate overconfident one in agricultural advisory: the overconfident system’s wrong predictions come with high-confidence labels that trigger wrong treatments. PlantSwarm targets the ECE column, not only F1. The architectural claim—that confidence-gated routing produces lower ECE independently of backbone—holds even if absolute F1 gains are modest.

Per-image routing beats distribution-level learned routing.

FrugalGPT knows “60% of images are easy” from training statistics. It cannot identify *which* image is easy. PlantSwarm’s κ signal and entropy gradient provide per-image uncertainty signals that learned cascade routers structurally cannot, which is why routing gains are expected to be largest on PlantDoc and LeafBench (high image-level difficulty variance) rather than PlantVillage (uniformly easy).

Entropy routing closes the theory-implementation gap.

The amortized inference framing (Eq. 1) requires minimizing posterior entropy per step. The κ signal approximates this linguistically; H_t and G_t operationalize it exactly. P4 tests whether the approximation is good enough—if $\rho < 0.30$, the Entropy-Gated Swarm is not merely a theoretical improvement but a practically necessary one.

Method	PlantVillage		PlantDoc		PlantWild		LeafBench		ECE↓	TPCP↓
	T3	T2	T3	T2	T3	T2	T3	T2		
Single VLM	—	—	—	—	—	—	—	—	—	—
MVPDR	—	—	—	—	—	—	—	—	—	—
Snap & Diagnose	—	—	—	—	—	—	—	—	—	—
SAGE (no KB)	—	—	—	—	—	—	—	—	—	—
SAGE+KB ($k=16$)	—	—	—	—	—	—	—	—	—	—
Chat Demeter	—	—	—	—	—	—	—	—	—	—
FrugalGPT	—	—	—	—	—	—	—	—	—	—
Fixed Chain	—	—	—	—	—	—	—	—	—	—
Fixed Chain + Ctx	—	—	—	—	—	—	—	—	—	—
3-Agent Swarm	—	—	—	—	—	—	—	—	—	—
PlantSwarm	—	—	—	—	—	—	—	—	0.3103	0.0
Entropy-Gated	—	—	—	—	—	—	—	—	—	—

Table 5: Main results: T3 (disease name) and T2 (pathogen class) Macro-F1, ECE, and TPCP. Numeric cells are generated from results/plantswarm_metrics.json by scripts/sync_latex_metrics.py. ECE is the primary calibration metric. Hard-split columns (PlantDoc, LeafBench) are most informative for routing differences.

Variant	PlantDoc		LeafBench	
	T3	ECE	T3	ECE
Fixed Chain	—	—	—	—
FC + Context	—	—	—	—
Free, No κ	—	—	—	—
Free, No Backtrack	—	—	—	—
3-Agent Swarm	—	—	—	—
PlantSwarm	—	—	—	—
Entropy-Gated	—	—	—	—

Table 6: Ablation on hard splits. Reading down isolates one component per column. If FC+Ctx closes the gap with PlantSwarm, context—not routing flexibility—is the primary driver.

Test	Expected	Result
P1 $\rho(L, H_{\bar{g}})$	+0.42+0.55	pending (—)
P2 ΔAcc (2nd—1st pass, Pa)	+9 F1	see traces
P3 $\text{acc}(\text{early})-\text{acc}(\text{extended})$	+12 F1	pending (—)
P4 $\rho(H^{(\text{dis})}, \kappa=L)$	+0.55+0.70	pending (—)

Table 7: Four falsifiable predictions. Weak P4 (<0.30) independently motivates Entropy-Gated Swarm.

Limitations. The routing policy is prompt-encoded, not learned—RL or preference optimization could improve edge-case calibration. Routing path agreement ($\sim 68\text{--}76\%$ across repeated runs for κ -gated; expected $\sim 80\%$ for entropy-gated, which is deterministic given weights and input) implies residual stochasticity. Coverage of 14 crops leaves some regional varieties underrepresented; LeafBench partially addresses this. PDS localizes difficulty to disease categories but not to specific visual features—qualitative trace analysis is needed

Mechanism	Expected	Result
Retrospective grounding ΔAcc (Pa)	+9 F1	—
Retrospective grounding ΔAcc (Sy)	+5 F1	—
Contradiction detection $P(\text{step}_j)$	0.72–0.80	—
Hedge propagation ρ	−0.38–0.52	—

Table 8: Context buffer mechanism validation.

Category	Type	PDS _{trace}	PDS(c)	Δacc
Viral (mosaic, TYLCV, CMD)	T2	$\approx +1.3$	—	−10–−15
Early fungal	T4+T2	$\approx +1.1$	—	−8–−12
Nutrient deficiency	T2	$\approx +0.9$	—	−7–−10
Bacterial blight	T2	$\approx +0.7$	—	−5–−8
Severe fungal	T2+T4	≈ -0.5	—	+5–+8
Powdery mildew	T1	≈ -0.3	—	+3–+5
Healthy	T4	≈ -0.8	—	+8–+12

Table 9: PDS by category. Δacc in F1 points. PDS(c) column populated from experiments. $\rho(\text{PDS}_{\text{trace}}, \Delta\text{acc}) \approx -0.65$.

to pinpoint confounders.

9 Conclusion

PlantSwarm demonstrates that the uniformity assumption baked into every prior agentic plant disease system—same compute, same confidence, same chain for every image—is both technically wasteful and agronomically harmful. Confidence-gated free routing, grounded in amortized posterior inference and operationalized at two levels (prompt κ and generation entropy), produces diagnostic trajectories that scale to image difficulty: short and confident on obvious cases, extended and self-correcting on genuinely ambiguous ones. The entropy field formally defines five

uncertainty quantities (Eqs. 2–7), enabling routing based on gradient and dispersion rather than scalar self-report. The Pathogen Difficulty Score in its entropy-native form (Eq. 13) turns routing traces into a pre-deployment audit tool grounded in model internals. Seven-variant factorial ablation isolates the contribution of each component. All traces, code, and PDS scripts are released (Liu et al., 2026).

Ethics Statement

PlantSwarm is a research prototype. Production deployment requires validation on local varieties not in our four benchmarks. ECE values are released per disease category so practitioners know exactly where system confidence is reliable. Treatment recommendations in DiagnosisAgent’s output are informational only; agronomic decisions must involve trained extension workers. PlantDoc and LeafBench include Sub-Saharan African and South Asian field conditions, ensuring PDS and calibration results reflect smallholder deployment contexts where accurate diagnosis has the highest potential impact.

Limitations

See Section 8. PlantSwarm is a research prototype not yet validated for production deployment.

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A Algorithms

B Full Agent Prompts

MorphologyAgent.

You are MorphologyAgent, a visual grounding specialist. Describe lesion shape (circular, angular, irregular), color (yellow halo, brown center, black margins), distribution (scattered, marginal, systemic), affected tissue (leaf, stem, fruit, root), necrosis pattern (tip burn, interveinal, whole leaf), and surface texture (powdery, water-soaked, dry, sporulating). DO NOT identify any disease or pathogen. Select ONE of [SymptomAgent, SeverityAgent]. Report confidence (high/medium/low) and use handoff tool.

SymptomAgent.

You are SymptomAgent. T1: classify symptom as Lesion/Blight/Mosaic/Wilt/Rot/Canker/Rust/Powdery Mildew. Read ALL prior messages. Assess whether pattern is pathognomonic or ambiguous. If confidence=low AND no prior backtrack: handoff to MorphologyAgent. If already

Algorithm 1 Entropy-Driven Free-Routing Swarm (PlantSwarm)

Require: Image X , agents A , $T_{\max}$

Ensure: $\hat{y}$, trace τ , $\mathcal{E}$

```

1:  $C_0 \leftarrow \langle (\text{USER}, X) \rangle$ ;  $\mathcal{E}_0 \leftarrow \emptyset$ ;  $a_1 \leftarrow \text{MO}$ ;  $t \leftarrow 1$ 
2: while  $t \leq T_{\max}$  do
3:    $(y^{(t)}, \{\log p_i^{(t)}\}) \leftarrow a_t(X, C_t)$   $\triangleright$  agent generation
4:   for each token  $i$  do
5:      $h_i^{(t)} \leftarrow -\sum_{v \in V} p(v|y_{<i}) \log p(v|y_{<i})$   $\triangleright$  Eq. 2
6:   end for
7:    $H_t \leftarrow \frac{1}{N} \sum_i h_i^{(t)}$ ;  $D_t \leftarrow \text{Var}(h^{(t)})$   $\triangleright$  Eqs. 3,6
8:    $C_{t+1} \leftarrow C_t \oplus (a_t, y^{(t)})$ ;  $\mathcal{E}_{t+1} \leftarrow \mathcal{E}_t \cup \{(H_t, h^{(t)})\}$ 
9:    $G_t \leftarrow \frac{1}{t-1} \sum_{k=2}^t (H_k - H_{k-1})$  if  $t > 1$  else 0  $\triangleright$  Eq. 5
10:  if  $G_t > \delta_1$  then  $a_{t+1} \leftarrow \text{MO}$   $\triangleright$  backtrack
11:  else if  $H_t < \delta_2$  then  $a_{t+1} \leftarrow \text{DI}$   $\triangleright$  early terminate
12:  else  $a_{t+1} \leftarrow \pi(a_t)$   $\triangleright$  forward
13:  end if
14:  if  $a_{t+1} = \text{DI}$  then return  $\hat{y}, \tau, \mathcal{E}$ 
15:  end if
16:   $t \leftarrow t + 1$ 
17: end while
```

backtracked: forward. If confidence=high: handoff to PathogenAgent. 995
996
997

PathogenAgent.

You are PathogenAgent. 999
T2: pathogen class 1000
(Fungal/Bacterial/Viral/Nutrient Deficiency/Pest Damage). T3: 1001
specific disease name (e.g. Cassava 1002
Bacterial Blight, TYLCV, Alternaria 1004
Leaf Spot). Focus on margin 1005
definition (Fungal= defined, 1006
Bacterial=angular/water-soaked), 1007
color patterns (Viral=mosaic, 1008
Nutrient=interveinal chlorosis), 1009
sporulation (Fungal diagnostic). 1010
Report confidence. If low AND 1011
no prior backtrack: handoff to 1012
MorphologyAgent. 1013

SeverityAgent.

You are SeverityAgent. T4: severity 1015
(Healthy/Early/Moderate/Severe). T5: 1016
crop species. For T4: estimate 1017
affected area, lesion stage, 1018
secondary infection, systemic vs 1019
local. T5 should be obvious--flag 1020

Algorithm 2 Fixed Sequential Chain (Baseline)

Require: X ; order $\mathcal{O}=[\text{MO}, \text{SY}, \text{PA}, \text{SE}, \text{DI}]$

Ensure: $\hat{y}, \tau, \Omega$

- 1: $C_0 \leftarrow \langle (\text{USER}, X) \rangle$; $\tau \leftarrow []$; $\Omega \leftarrow 0$
 - 2: **for each** a in $\mathcal{O}$ **do**
 - 3: $(y^{(t)}, \kappa_t, _) \leftarrow a(X, C_t) \triangleright \kappa_t$ computed,
 never used
 - 4: $\Omega \leftarrow \Omega + \text{tokens}(y^{(t)})$; $C_{t+1} \leftarrow C_t \oplus$
 $(a, y^{(t)}, \kappa_t)$
 - 5: **end for**
 - 6: **return** $\text{DI}(C_t, \text{aggregate}), \tau, \Omega$
-

if uncertain. If confidence=low
AND no prior backtrack: handoff to
MorphologyAgent. Otherwise handoff
to DiagnosisAgent.

DiagnosisAgent.

You are DiagnosisAgent. Read
ALL prior messages. Prefer
later, more-informed predictions
when resolving contradictions.
Output JSON: symptom_complex
(T1), pathogen_class (T2),
disease_name (T3), severity
(T4), crop_species (T5),
confidence_t1..t5, prediction_entropy
(0.0-1.0), contradiction_resolved
(bool), backtrack_count,
treatment_recommendation
(informational only), path_summary.
Then output TERMINATE.

C Benchmark Construction

PlantVillage. 5,000 images from [Hughes and Salathé \(2015\)](#), stratified by crop $\times$ disease. Color format only.

PlantDoc. Full 2,569 images from [Singh et al. \(2020\)](#). 13 crops, 27 disease classes, web-scraped with variable backgrounds.

PlantWild. 4,000 sampled from [Wei et al. \(2024a\)](#). Largest in-the-wild benchmark with text disease descriptions.

LeafBench. 3,500 from [Quoc et al. \(2026\)](#). 64 disease categories including rare and regional varieties. Primary PDS benchmark.

Stratification. All splits balanced by crop $\times$ pathogen class $\times$ severity $\times$ image source (controlled/field).

D Dataset licences

Table 10 summarises hosting and distribution terms for the four benchmarks used in evaluation.

E Entropy Field Implementation

Extraction. logprobs=True passed to vLLM client. Per-token entropy: $h_i = -\log p_i$ (single-sample estimate, exact under greedy decoding). $H_t^{(\text{dis})}$ extracted from the token span of PathogenAgent’s disease_name JSON field.

Hyperparameter sensitivity. $\delta_1 \in \{0.10, 0.15, 0.20, 0.25\}$ and δ_2 tuned on 500-image held-out set; full sensitivity table reported here after experiments.

Cross-model. Entropy values comparable across Qwen3-VL-8B, Phi-4-multimodal, Qwen2.5-VL-7B because all use same temperature and the entropy field is read from the generation distribution, not logit scaling.

F Budget Sensitivity**G Path Clustering**

Paths encoded as strings over $\{\text{Mo}, \text{Sy}, \text{Pa}, \text{Se}, \text{Di}\}$. Pairwise Levenshtein distances feed hierarchical agglomerative clustering (Ward linkage); cluster count maximizes silhouette score. Per-cluster disease-category distribution and quality statistics characterize which image types trigger which routing strategies.

H Hedge Lexicon

Seed terms: *possibly, unclear, hard to tell, uncertain, approximately, may, might, could, appears to be, cannot rule out*. Co-occurring n-grams extracted within 5-token window, scored by PMI. Top-200 PMI candidates manually reviewed. Validated against BioScope ([Vincze et al., 2008](#)) and CoNLL 2010 Shared Task.

Table 10: Dataset licence summary for the **four** evaluation benchmarks: hosting, distribution rights, and primary citation.

Dataset	Licence	Notes & Citation	Link
PlantVillage	CC BY-SA 3.0	Attribution and ShareAlike required; derivatives must carry the same licence. Hughes & Salathé, arXiv:1511.08060, 2015; Mohanty et al., <i>Front. Plant Sci.</i> 2016.	Link
PlantDoc	CC BY 4.0	Attribution required. Singh et al., <i>ACM IKDD CoDS & COMAD</i> , 2020. DOI: 10.1145/3371158.3371196.	Link
PlantWild v2	CC BY-NC-ND 4.0	No commercial use; no derivatives permitted. Wei et al., <i>ACM MM</i> , 2024. DOI: 10.1145/3664647.3680599.	Link
LeafBench	CC BY 4.0	Released on Hugging Face as enalis/LeafNet; $\sim 70\%$ of data is public (training split only). Quoc et al., arXiv:2602.13662, 2026.	Link

β	T3 F1	T2 F1	ECE	Mean L	TPCP
0 (no BT)	—	—	—	—	—
1 (default)	—	—	—	—	—
2 (two BT)	—	—	—	—	—

Table 11: Backtrack budget sensitivity on PlantDoc (1k subset). Expected: $\beta=1$ optimal; $\beta=2$ marginal gain at $\sim 15\%$ token overhead.